

Enterprise Data Management Analysis for Third Star Financial Services

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Final

Third Star Financial Services is a global alternative banking provider serving over 15,000 locations and an expanding online platform. Historically, the organization has grown through the acquisition of smaller financial institutions, leading to a disjointed IT and data landscape. The lack of integration across systems has resulted in poor data quality, limited analytic capabilities, and operational inefficiencies.

To support growth and transformation into a digital-first financial services provider, Third Star must implement a modern *enterprise data management* (EDM) strategy. This report evaluates the current state of Third Star's data environment, recommends technologies and governance improvements, and outlines a clear implementation plan aligned with EDM standards.

I. Current State Assessment

Summary of Current Architecture

Third Star's IT infrastructure is fragmented, with Oracle and Microsoft SQL Server databases operating across a mix of Windows, UNIX, and Linux systems. Each acquired business unit supports its own systems, which are not integrated with the main online platform launched in 2009. These data silos have led to inconsistent and redundant data.

The company has attempted data warehousing in the past, but these efforts have been LOB specific, not delivering enterprise-wide benefits. Cognos is used for operational reporting but lacks the capability for advanced analytics.

Summary of Needs

Effective data management needs a coordinated strategy across 11 key knowledge areas (DAMA, 2017). Third Star lacks development in many foundational areas, including:

- Data Governance: No formal governance policies or ownership roles exist.
- Data Architecture: There is no standardized blueprint aligning data design with business goals.
- Master Data Management (MDM): There is no single source of truth for customers, agents, or products.
- Metadata Management: A centralized metadata repository or business glossary is absent.
- Data Quality: Redundancy and inconsistency are rampant due to a lack of validation rules or cleansing procedures.

Operational Impact

Poor data quality has led to service delays, duplicate records, and limited insight into customer behavior. The company's market share has dropped from 24% to 18.1% in part due to untrustworthy reporting and poor customer experiences. Without changes, future innovations, such as mobile banking or real-time transaction reporting, will be hindered.

II. Data Model

To establish a strong data foundation, conceptual and logical models were developed focusing on the customer subject area.

Conceptual Model Overview

The conceptual model includes the following entities:

- Customer: Central record for individuals using Third Star's services.

- Address: Linked to Customer; allows multiple address types (e.g., mailing, billing).
- Transaction: Captures customer transaction details for reporting and compliance.
- Agent: Employees or representatives who assist customers.
- Region: Defines geographical grouping for agents and customers.

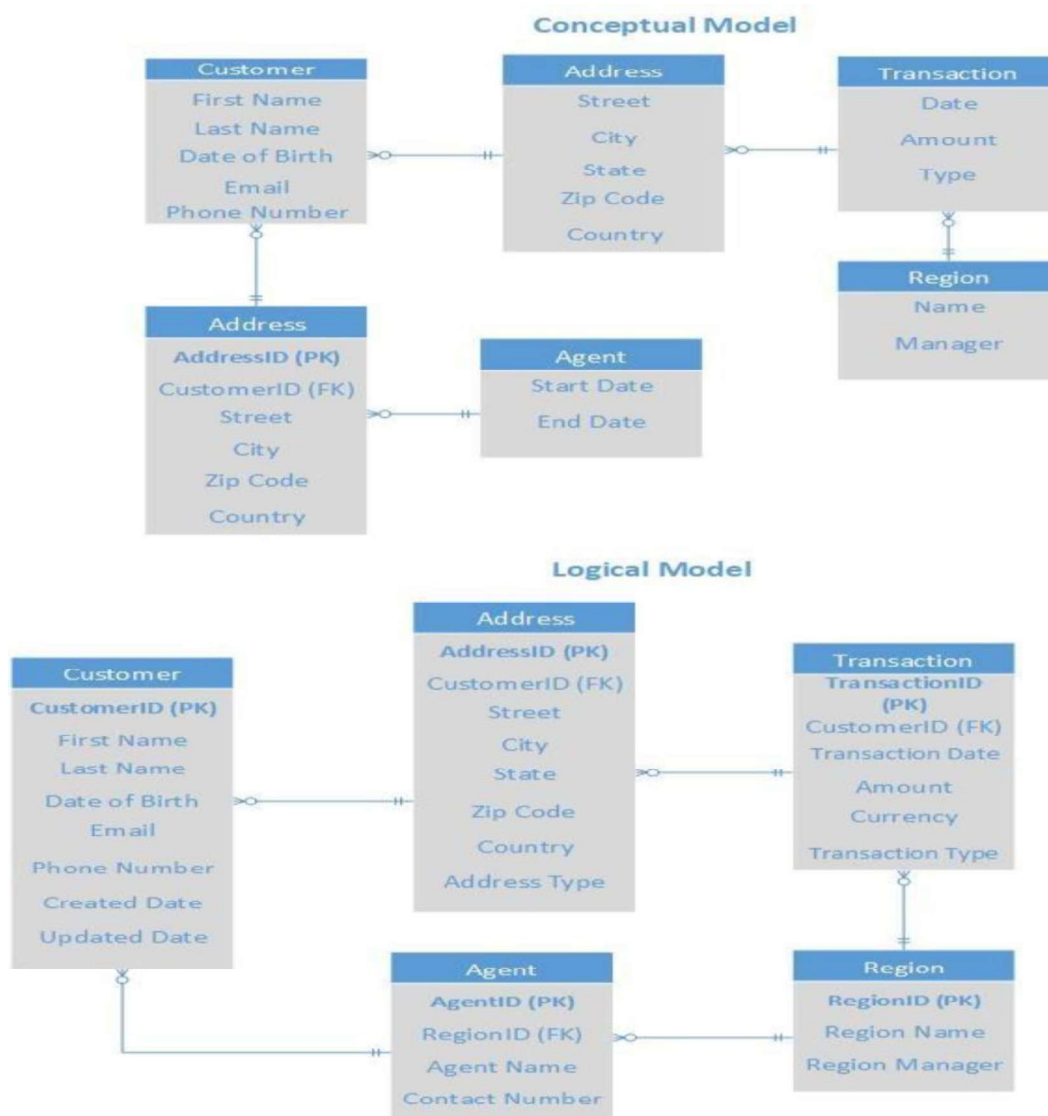
This model reflects the business requirements and connects business strategy with data architecture (DAMA, 2017).

Logical Model Overview

The logical model introduces:

- Primary Keys (CustomerID, TransactionID)
- Foreign Keys (CustomerID in Address and Transaction)
- Operational attributes such as Created Date, Updated Date, Address Type, etc.

These structures support future physical database design and enable data validation, consistency, and integration (DAMA, 2017).



III. Recommendations

Technology Recommendations

Master Data Management (MDM)

An MDM platform, such as IBM InfoSphere, centralizes and standardizes critical data entities, such as customer and product data, resolving inconsistencies across isolated systems and improving data quality, consistency, and trustworthiness. MDM enables enhanced analytics and reporting by ensuring reliable master data for accurate cross-system reporting, customer

segmentation, and strategic insights. It also simplifies regulatory compliance by ensuring data accuracy, lineage, and auditability (DAMA, 2017).

Implementing an MDM solution is complex, requiring significant data integration efforts, system mapping, and stakeholder alignment to define golden records. Additionally, business units accustomed to local data definitions may resist centralized standards, necessitating strong data governance and change management strategies (DAMA, 2017).

Data Quality Management (DQM)

DQM tools, like Talend Data Quality, automate data profiling, cleansing, and monitoring, efficiently identifying, correcting, and preventing data errors. High quality data is foundational to operational processes, customer experience, and predictive analytics. Additionally, these tools reduce compliance risk by ensuring data accuracy, completeness, and timeliness for regulatory reporting (DAMA, 2017).

DQM tools require upfront licensing costs and staff training to maximize effectiveness. Tools alone do not resolve data quality issues without clear business rules and data stewardship to guide profiling, validation, and cleansing activities (DAMA, 2017).

Managment Recommendations

- Establish a data governance framework
- Implement a data quality management program
- Align data architecture

Combined Impact

Together, MDM and DQM will enable Third Star to merge systems, improve data reliability, and better serve both internal and external stakeholders. A governance framework ensures long-term effectiveness of these technologies by enforcing rules and roles around data use (DAMA, 2017). Clear data definitions and business rules must be in place for tools to function as intended, and tools will support those rules in turn.

Budgeting restrictions should also be considered when selecting technologies and tools. Estimated cost-to-value returns should be calculated and approved before final tool selections are made.

IV. Implementation Plan

Phased Implementation

- Phase 1: Planning and Governance Setup
 - Establish a cross-functional data governance council.
 - Define data domains, stewardship roles, and approved business rules.
 - Select specific MDM and DQM vendors
- Phase 2: Tool Installation and Configuration
 - Install MDM and DQM tools.
 - Integrate with core systems.
 - Configure validation rules and golden record criteria.
- Phase 3: Data Profiling, Cleansing, and Mastering
 - Profile and cleanse data sets.
 - Apply MDM rules to master entities.
 - Monitor quality metrics.

- Phase 4: Training, Policy Finalization, and Rollout
 - Train users.
 - Finalize data quality policies.
 - Deploy dashboards and monitoring.

Additional Requirements

- Inventory existing data sources and systems.
- Complete a data quality assessment.
- Define ownership and stewardship roles.
- Clarify system integration points for analytic tools.

Management Attention

Executive sponsorship is crucial. Leadership must support policy enforcement, allocate resources, and promote adoption through communication and training. The proposed strategy includes governance council formation and stakeholder engagement to meet these needs (DAMA, 2017).

Technical Expertise Required

- Data Architect: Integration design
- ETL Developers: Data movement
- DBAs: Performance and security
- Data Stewards and Analysis: Ongoing maintenance
- Vendor Support: Short term implementation assistance

V. Financial and Ethical Standards

Restrictions to Protect Financial Data

- Data Encryption: Both at rest and in transit
- Role-Based Access Control (RBAC): Enforce principle of least privilege
- Audit Trails: Track access and changes
- Network Segmentation: Isolate sensitive systems
- Vendor Contracts: All vendor relationships involving financial data must include contractual clauses mandating compliance with relevant data privacy laws (e.g., PCI DSS) and security standards.

Vendor contracts need to be detailed when specifying the above measures; they should also:

- Require strong encryption, access control, and vulnerability management measures.
- Specify timelines and procedures for breach notification and cooperation in investigations.
- Grant audit rights for ongoing compliance monitoring.
- Include indemnification clauses to hold vendors financially responsible for security incidents.
- Require secure data return or destruction upon contract termination, with written confirmation.

Such provisions ensure vendors are held to the same security and compliance obligations as Third Star, reducing risk and meeting legal requirements (Aaron Hall, 2023; Morris James LLP, 2017; NAHB, 2020; UpGuard, 2022).

Ethical Considerations

- Privacy vs. Insight Tradeoffs: Use anonymization where possible
- Bias in Financial Modeling: Audit algorithms for fairness
- Informed Consent and Transparency: Clear customer consent and opt-in policies
- Access to Restrictions Based on Ethical Need: Limit combined access to sensitive personal and financial data
- Whistleblower and Escalation Mechanisms: Allow internal reporting of ethical concerns

VI. Conclusion

Implementing a modern enterprise data management strategy, centered on MDM, DQM, and formal governance, will enable Third Star to improve operational efficiency, rebuild customer trust, and prepare for digital innovation. By aligning tools, people, and processes, Third Star will gain a competitive edge in the financial services market. Future recommendations include:

- Exploring metadata management tools
- Scaling to include predictive analytics and AI models
- Continuous review of governance policies and ethical practices

By combining technology with ethical responsibility and organizational alignment, Third Star can build a sustainable data-driven future.

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